

WSL 1 and WSL 2 is shown in Figure 7. WSL 2 enables virtualisation through the native virtualisation technology of Windows OS and Hyper-V. It does create a virtual machine behind the scenes, but the creation is done by WSL 2 in a highly optimised manner. It ensures less resource utilisation by allocating resources dynamically as and when required by the virtual machine. Also, Windows automatically switches off inactive virtual machines. This contributes to reducing the time to start the virtual machine.

Currently, WSL 2 is available by default in Windows 10 Version 2004. However, users of Windows 10 Version 1903 (build number 18362.1049+) and Windows 10 Version 1909 (build number 18363.1049+) can manually upgrade to WSL 2. Follow the steps given in the next section to do so. Windows also allows both versions of WSL to run in parallel. We can run Ubuntu distro on a Windows machine with WSL 2 and Kali Linux with WSL 1.

Install WSL 2

To install WSL 2 on Windows 10 Version 1903 or Windows 10 Version 1909, follow the steps given below.

- Before installing WSL 2, you need to make sure that your OS build number is appropriate. To check this, do the following:
 - Press the Windows key + R and enter *winver*.
 - Click on *OK*. This shows the OS build information in a new window.
- If your OS build number conforms to the above-stated requirements, then skip this step. Otherwise, you need to update your OS to the latest version. This can be done by checking for available Windows updates and installing them. The steps for the same are given below:
 - Click on the *Start* button and open *Settings*. Then go to *Update and Security*.
 - Click on *Check for updates* and install all the available updates.

Feature	WSL1	WSL2
Integration between Windows and Linux	Yes	Yes
Fast boot times	Yes	Yes
Small resource footprint	Yes	Yes
Runs with current versions of VMware and VirtualBox	Yes	Yes
Managed virtual machine	No	Yes
Full Linux kernel	No	Yes
Full system call compatibility	No	Yes
Performance across OS file systems	Yes	No

Figure 7: A comparison between the features of WSL 1 and WSL 2

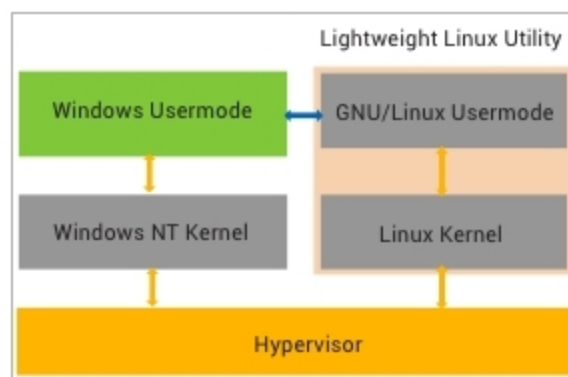


Figure 8: WSL 2 architecture

- Now, we need to enable the WSL and virtual machine platform features of Windows. For this, run the following commands in Windows PowerShell with administrative privileges:

```
$dism.exe /online /enable-feature /
featurename:Microsoft-Windows-Subsystem-
Linux /all /norestart
```

```
$dism.exe /online /enable-feature /fea
turename:VirtualMachinePlatform /all /
norestart
```

- Download the latest Linux kernel update package from https://wslstorestorage.blob.core.windows.net/wslblob/wsl_update_x64.msi for x64 machines and from https://wslstorestorage.blob.core.windows.net/wslblob/wsl_update_arm64.msi for ARM64 machines. The command `systeminfo | find 'System Type'` shows the architecture used in your system. This will help you select the appropriate package.
- Run the package downloaded in the previous step.

- Restart the machine to finish WSL 2 installation.

To know more about the nuances of each step or about troubleshooting errors, visit <https://docs.microsoft.com/en-us/windows/wsl/install-win10>.

Docker Desktop for Windows

Docker Desktop is an application that can be used on Windows for the development of containerised applications. It includes Docker Engine, Docker CLI client, Docker Compose, Notary, Kubernetes and Credential Helper. Using Docker Desktop, we can build and publish Docker images, run them in Docker containers, get information about the resource utilisation of containers, and much more, all through a user-friendly GUI. For experienced users, it also provides a CLI client. Docker Desktop also allows us to access the Docker Hub repository from its interface. Thus, we can download images from public repositories and private repositories.

But why is using Docker on a Windows machine difficult and not as seamless as in the case of Linux machines? What are the mechanisms that make this possible? The limitation of Docker is that it can only run on the Linux kernel since type-1 and type-2 hypervisors use hardware-level virtualisation, whereas containers use kernel-based virtualisation. To run Docker on Windows, we need virtualisation software that can emulate a Linux kernel. Docker Desktop utilises Microsoft's Hyper-V hypervisor, which is a native hypervisor capable of creating virtual machines on x86-64 machines running the Windows operating system. Figure 8 shows how any Linux distribution can run on a full Linux kernel made available by WSL 2.

Docker Desktop integrates easily with WSL 2 and can be run with the WSL 2 backend, as it takes advantage of the full Linux kernel and Linux system call support provided by WSL 2. WSL 2 allows users to use Linux